ASSIGNMENT # 04

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FARWA LAEEQ (0027)

SUBJECT : AI

SECTION : E2

SUBMITTED TO: Ma'am Hina Tahir

CamScanner

ASSIGNMEN-04

NADIA AKRAM (0051)
FARWA LAEEQ (0027)

Question no #1

NAIVE BAYES:-

SAMPLE:

Birth	Fly	in water	legs	class
yes	No	Yes	No	?

Total = 20

P(Mammals) = 7/20

P(Non-Mammals) = 13/20

For Mammals:-

$$P(M/B,F,W,L) = P(M) \times P(Birth, Yes/M) \times P(No\ Fly/M) \times P(in\ water/M) \times P(No\ legs/M)$$

$$= 7/20 \times 6/7 \times 6/7 \times 2/7 \times 2/7$$

$$= 0.35 \times 0.86 \times 0.86 \times 0.26 \times 0.26$$

$$P(M/B, NF, W, NL) = 0.02$$

For Non-Mammals:-

$$P(NM/B, NF, W, NL) = P(NM) \times P(B/NM) \times P(NF/NM) \times P(W/NM) \times P(NLegs/NM)$$

$13/20 \times 1/13 \times 10/13 \times 3/13 \times 4/13$

$= 0.65 \times 0.077 \times 0.77 \times 0.23 \times 0.31$

$P(NM/B,F,N,L) = 0.0027$

As,

$P(Mammals/B,F,N,L) = 0.02$

$P(Non\text{-}Mammals/B,F,N,L) = 0.0027$

So,

Probability of Mammals is more than non-Mammals ($0.02 > 0.0027$)

Therefore,

Sample belongs to **Mammals**.

Question no:- 2)

Perceptron

$w_1 = 0$, $w_2 = 0$, $b = 1$, $\alpha = 1$

Iteration 1:-

Sample	x_1	x_2	Target	Σ	f(Activation)	b	w_1	w_2	status
1	2	1	1	1	1	1	0	0	No change
2	-1	-3	-1	1	1	0	1	3	change
3	1	-1	-1	-2	-1	0	1	3	No change

1) $x_1w_1 + x_2w_2 + b$

$= 2(0) + 1(0) = 1$

$= 1 \Rightarrow f(1) > 0 = 1$

2) $-1(0) + (-3)(0) + 1 \Rightarrow 0+0+1 = 1 > 0$

$= 1$

3) $(1)(1) + (-1)(3) + 0 \Rightarrow 1-3+0 = -2 < 0$

$= -1$

$\because$ net $> 0 \rightarrow 1$

else -1

$\Delta w_1 = \alpha \times$ target $\times x_1$

$= (1)(-1)(-1) \Rightarrow 1$

$w_1 = 1$

$\Delta w_2 = 1 \times (-1) \times (-3) \Rightarrow 3$

$w_2 = 0 + 3 = 3$

New bias = old + α x target

New bias = 1 + (1 x -1) = 1 - 1 = 0

Iteration # 2

Sample	x_1	x_2	target	$\sum$	Activation	b	w_1	w_2	Status
1	2	1	1	5	1	0	1	3	No change
2	-1	-3	-1	-10	-1	0	1	3	No Change
3	1	-1	-1	-2	-1	0	1	3	No Chang

$w_1 = 1$, $w_2 = 3$, $b = 0$

1) $2(1) + (1)(3) + 0$

$= 2 + 3 = 5 > 0 \Rightarrow 1$

2) $(-1)(1) + (-3)(3) + 0$

$= (-1 - 9) + 0 \Rightarrow -10 < 0 \Rightarrow -1$

3) $(1)(1) + (-1)(3) + 0$

$= (1 - 3) + 0 \Rightarrow -2 < 0 = -1$

All sample status are same so, we stop iterations.

Question no:-3

Movie	Genre	Explosions	Laughter	Calculation	Distance
M1	Action	10	2	$\sqrt{(6-10)^2+(4-2)^2}$	4.47
M2	Comedy	5	8	$\sqrt{(6-5)^2+(4-8)^2}$	4.12
M3	Action	8	3	$\sqrt{(6-8)^2+(4-3)^2}$	2.24
M4	Comedy	3	7	$\sqrt{(6-3)^2+(4-7)^2}$	4.24
M5	Action	12	1	$\sqrt{(6-12)^2+(4-1)^2}$	6.71
M6	Comedy	4	6	$\sqrt{(6-4)^2+(4-6)^2}$	2.83

SORTING IN ORDER (ASCENDING)

MOVIE	DISTANCE	GENRE
M3	2.24	Action
M6	2.83	Comedy
M2	4.12	Comedy
M4	4.24	Comedy
M1	4.47	Action
M5	6.71	Action

CONCLUSION:-

K = 3

Prediction:-

M3 → Action

M6 → Comedy

M2 → Comedy

It belongs to Comedy.

K=5

M 3 → Action
M6 → Comedy
M2 → Comedy
M4 → Comedy
M1 → Action

Prediction:- As 2 votes for Action & 3 for Comedy
So, it belongs to **Comedy**

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Question no:-5

Euclidean Formula: $\sqrt{(Y_2 - X_1)^2 + (Y_2 - Y_1)^2}$

Customer	Annual Income	Spending score	X_2 10,000	Y_2 60	X_2 80,000	Y_2 70	Cluster
C1	40,000	60	0		40000		1
C2	60,000	40	20000.01		2000002		1
C3	80,000	70	40000		0		2
C4	20,000	5	20000		60000		1
C5	120,000	90	80000		40000		2
C6	100,000	60	60000		20000		2

Cluster-1 : {C1, C2, C4}

Cluster-2 : {C3, C5, C6}

Centroid 1 :- $\left\{ \frac{40000 + 60000 + 20000}{3}, \frac{60+40+5}{3} \right\}$

$= (40,000, 35)$

Centroid 2 :- $\left\{ \frac{80000 + 120000 + 10000}{3}, \frac{70+90+60}{3} \right\}$

$= (100000, 13.33)$

			C1	C2		
Customer	A.Income	S.Score	40,000 | 35	100000 | 73.33	CLUSTER	NEW CLUSTER
C1	40,000	60	25	600000	1	1
C2	60,000	40	2000000	40000	1	1
C3	80,000	70	40000	20000	2	2
C4	20,000	5	20000	80000	1	1
C5	120,000	90	80000	20000	2	2
C6	100000	60	60000	1333	2	2

CONCLUSION :-

CLUSTER 1:- (Low Income / Spending)

Customer = C1, C2, C4

Centroid = (40000, 35)

CLUSTER 2:- (HIGH Income / spending)

Customer = C3, C5, C6

Centroid = (100000, 73.33)